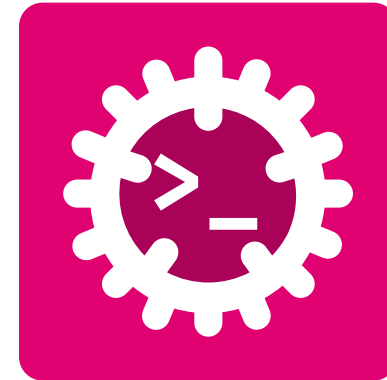


# Advanced Programming

Degub // Testing // Documentation



---

Silvan Zahno

University of Applied Sciences Western Switzerland - HES-SO Valais Wallis  
Systems Engineering  
Infotronics  
2026-08-07 - v0.1.0



## Contents

|                                         |    |
|-----------------------------------------|----|
| Debug / Testing / Documentation .....   | 3  |
| Ensuring correctness continuously ..... | 4  |
| Testing .....                           | 5  |
| Unit tests .....                        | 6  |
| Integration & Doc tests .....           | 7  |
| Assertions & expected failures .....    | 8  |
| Code Quality & CI .....                 | 9  |
| fmt, check, clippy .....                | 10 |
| Continuous Integration .....            | 11 |
| Handling Failures .....                 | 13 |
| Handling failures: the toolbox .....    | 14 |
| Handling failures: in practice .....    | 15 |
| Exercise - Handling Failures .....      | 16 |
| Debugging .....                         | 17 |
| Print & inspect .....                   | 18 |



|                                                               |    |
|---------------------------------------------------------------|----|
| Logging .....                                                 | 19 |
| Debugging with Breakpoints (in Zed / VS Code) .....           | 21 |
| Profiling .....                                               | 22 |
| Benchmark & profile <b>Measure before you optimise.</b> ..... | 23 |
| Mystery Program 1 .....                                       | 25 |
| Mystery Program 2 .....                                       | 26 |
| Mystery Program 3 .....                                       | 27 |
| Mystery Program 4 .....                                       | 28 |
| Mystery Program 5 .....                                       | 29 |
| Documentation .....                                           | 30 |
| RustDoc comments .....                                        | 31 |
| Doc comments: how they work .....                             | 32 |
| Glossary & Bibliography .....                                 | 33 |



# **Debug / Testing / Documentation**

---

Ship code that works - and prove it



The safety net of a healthy Rust project - each layer catches a different class of mistake.

- **Unit tests** - `#[test]` functions, run by `cargo test`
- **Integration & doc tests** - the `tests/` folder and runnable examples in docs
- **cargo check / clippy** - fast type-check and lints
- **cargo fmt** - one canonical style
- **Error handling** - `Result` / `Option`, no surprise crashes
- **Logging & debugging** - see what the program actually did
- **Profiling** - measure before you optimise
- **CI/CD** - run **all** of the above automatically on every push



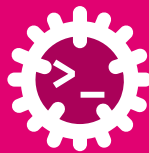
The goal: catch a mistake **as early and as cheaply** as possible - ideally at compile time.



# Testing

---

Prove the code does what you claim



A **unit test** checks **one** piece of code in **isolation** - including the **edge cases**.

```
// the code under test
pub fn parse_temperature(s: &str) → Option<i32> {
    s.trim().trim_end_matches("°C").parse().ok()
}

#[cfg(test)]                // compiled only for `cargo test`
mod tests {                 // a *submodule* for the tests
    use super::*;           // bring the parent module into scope

    #[test]                  // a *unit test* function
    fn plain_number() {
        assert_eq!(parse_temperature("20"), Some(20));
    }

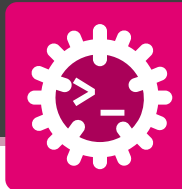
    #[test]
    fn strips_unit_and_spaces() {
        assert_eq!(parse_temperature(" 20°C "), Some(20));
    }

    #[test]
    fn rejects_garbage() {
        assert_eq!(parse_temperature("hot"), None);
    }
}
```

- lives **next to the code**, in `#[cfg(test)] mod tests`
- `#[test]` marks each test function
- the rhythm: **arrange** ⇒ **act** ⇒ **assert**
- test the **happy path**, the **edge**, and the **failure**
- run them all with `cargo test`



A failure prints the **file, line** and **both values** - it points right at the bug.



The **same** function, tested two more ways - both run under cargo test.

## Integration Tests

files in `tests/`, exercise the crate **from outside**

- only public API is visible

```
// tests/temperature.rs
use weather::parse_temperature;

#[test]
fn reads_celsius() {
    assert_eq!(parse_temperature("20°C"), Some(20));
    assert_eq!(parse_temperature("nope"), None);
}
```

## Doc tests

Code in `///` examples is **compiled and run** ⇒  
documentation that **can't go stale**

```
/// Parses a temperature like `20°C`. DOC test
///
/// # Examples
/// ```
/// use weather::parse_temperature;
/// assert_eq!(parse_temperature("20°C"), Some(20));
/// ```
pub fn parse_temperature(s: &str) → Option<i32> {
    // ...
}
```





An assertion says what **must** be true - and sometimes the **correct** behaviour is to **panic**.

```
// °F → °C, rejecting impossible input
fn f_to_c(f: f64) → f64 {
    assert!(f ≥ -459.67, "below absolute zero");
    (f - 32.0) * 5.0 / 9.0
}

#[test]
fn freezing_and_boiling() {
    assert_eq!(f_to_c(32.0), 0.0); // = (both shown on fail)
    assert_ne!(f_to_c(212.0), 0.0); // ≠ (it is 100.0)
}

#[test]
#[should_panic(expected = "below absolute zero")]
fn rejects_impossible() { f_to_c(-500.0); } // pass if it panics

#[test]
fn parses_ok() → Result<(), String> {
    let t = parse_temperature("20°C").ok_or("bad")?;
    assert_eq!(t, 20); // a test may return Result
    Ok(())
}
```

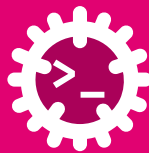
- `assert_eq!` / `assert_ne!` print **both values** on failure
- `assert!(cond, "msg")` - assert a bool, with a **message**
- `#[should_panic]` - the test **passes only if it panics**
- a test may **return Result** - ? inside, an `Err` fails it
- filter: `cargo test freezing` runs all function with the word `freezing` in the function name.  
⇒ Here only `freezing_and_boiling()`



## Code Quality & CI

---

Automate the boring checks



Three commands that catch **most** mistakes before a single test runs.

```
cargo fmt      # one canonical style
cargo check    # type-check only, no binary
cargo clippy    # ~750 lints, catches foot-guns
# auto-fix where possible
cargo fmt --all
cargo clippy --fix
```

Rust **fmt** can be configured with `rustfmt.toml` - but the **default** is good enough for most projects. Just run it **on save** in your editor and forget about it.

```
edition = "2024"
tab_spaces = 2
max_width = 160
reorder_imports = true
```

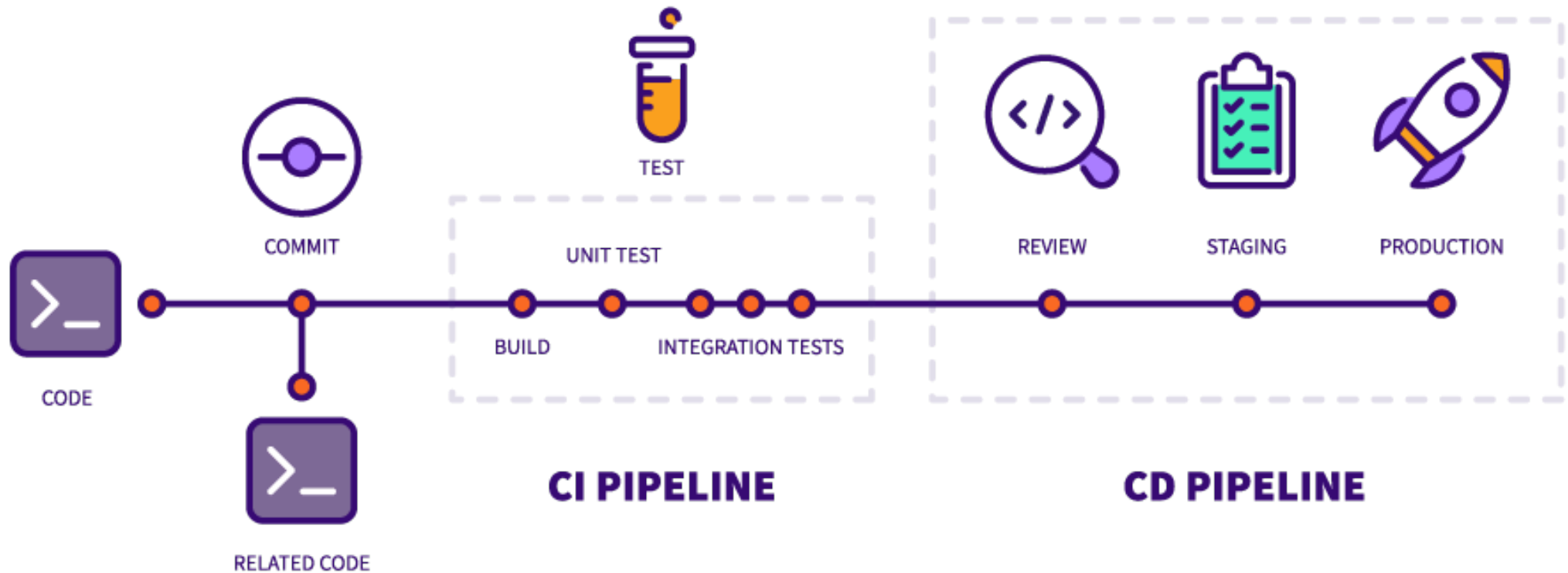
- **fmt** - rustfmt; run it **on save** and end style arguments
- **check** - compiler **front-end** only → fast feedback loop
- **clippy** - idiomatic-Rust **linter**; explains **why** and suggests a fix



In **CI**, make warnings fatal: `cargo clippy -- -D warnings`.



Run every check **automatically** on each **push / pull-request**





```
# .github/workflows/ci.yml
on: [push, pull_request]
jobs:
  fmt:
    - run: cargo fmt --all -- --check
  clippy:
    - run: cargo clippy -- -D warnings
  test:
    - run: cargo test --verbose
  build:
    - run: cargo build --verbose
```

- the **same** commands you run locally - now enforced for **everyone**
- four parallel jobs: **fmt** · **clippy** · **test** · **build**
- on **every push & PR** to main



Green check = formatted, lint-clean, builds, **all tests pass.**

The screenshot shows the GitHub Actions interface for a workflow named `ci.yml` triggered by a push to the `main` branch. The status is **Success**. The summary shows a total duration of **1m 38s**. A list of jobs on the left includes Test Suite, Rustfmt, Clippy, Build Check, and cross-platform builds for Ubuntu, Windows, and macOS, all marked with green checkmarks. The right panel shows a detailed view of the `ci.yml` workflow, listing the duration of each job: Test Suite (50s), Rustfmt (8s), Clippy (24s), and Build Check (50s). Below this, a 'Matrix: Cross-platform build' section indicates that **3 jobs completed**.

**`unwrap()` was a mistake**



## Handling Failures

---

When things go wrong, gracefully



`Result<T, E>` and `Option<T>` make failure **part of the type** (see **Imperative Programming**).

| Method                           | What it does                                         | Use when                                         |
|----------------------------------|------------------------------------------------------|--------------------------------------------------|
| <b>Propagate / handle</b>        |                                                      |                                                  |
| <code>?</code>                   | hands the <code>Err/None</code> to the <b>caller</b> | the caller should decide                         |
| <code>match / if let</code>      | branch on each case <b>explicitly</b>                | you handle both paths here                       |
| <b>Fallback - never panics</b>   |                                                      |                                                  |
| <code>unwrap_or(x)</code>        | the value, or <code>x</code>                         | you have a cheap default                         |
| <code>unwrap_or_else(f)</code>   | the value, or <code>f()</code>                       | the default is <b>expensive</b> to build         |
| <code>unwrap_or_default()</code> | the value, or <code>T::default()</code>              | <code>0</code> / <code>""</code> / empty is fine |
| <b>Panic - give up</b>           |                                                      |                                                  |
| <code>unwrap()</code>            | the value, or <b>panic</b>                           | tests / “can’t happen”                           |
| <code>expect("msg")</code>       | the value, or <b>panic w/ msg</b>                    | document the invariant                           |
| <code>panic!()</code>            | <b>abort</b> the thread                              | unrecoverable bug                                |



```
use std::fs;
fn load_port(path: &str) → Option<u16> {
    // ? propagates None upward if read fails
    let text: String = fs::read_to_string(path).ok()?;
    // ? propagates None upward if parsing fails
    let port: u16 = text.trim().parse().ok()?;
    Some(port)
}

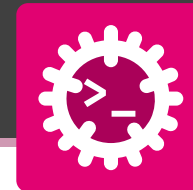
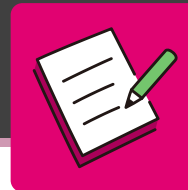
fn main() {
    // 1. Fallback - never panics: use 8080 if loading fails
    let port: u16 = load_port("app.toml").unwrap_or(8080);
    // 2. Handle both arms explicitly: success and failure
    if let Some(p) = load_port("app.toml") {
        println!("listening on {p}");
    } else {
        eprintln!("config problem, using defaults");
    }
    // 3. Fallback with a computed/expensive default
    let port: u16 = load_port("app.toml").unwrap_or_else(|| {
        eprintln!("falling back to default port");
        8080
    });
}
```

```
// 4. Fallback to the type's default value
let port: u16 = load_port("app.toml").unwrap_or_default();
// 5. Last resort in a prototype: panic with a message if
// config is missing/invalid
let port: u16 = load_port("app.toml").expect("config
required");
// 6. Unrecoverable state: abort explicitly
if port == 0 {
    panic!("port 0 is never valid");
}
}
```

## In the example

- ? - bubble the error **up**
- unwrap\_or - safe default 8080
- if let / match - handle **both** arms
- expect - **panic** w/ message
- panic! - **abort** the thread





The code is given. On each `// TODO` line, add the **right method** from the failure toolbox - each line uses a **different** one (`?`, `unwrap_or`, `unwrap_or_else`, `unwrap_or_default`, `expect`).

```
use std::num::ParseIntError;

// parse_port returns Result<u16, ParseIntError>
fn parse_port(s: &str) → Result<u16, ParseIntError> {
    s.trim().parse()
}

// TODO 1: propagate the Err to the caller (one operator)
fn sum_ports(a: &str, b: &str)
    → Result<u16, ParseIntError> {
    let x = parse_port(a)/* ? */; // → u16
    let y = parse_port(b)/* ? */; // → u16
    Ok(x + y)
}
```

```
fn main() {
    // TODO 2: use 8080 when parsing fails (cheap default)
    let p1: u16 = parse_port("9000")/* ? */;

    // TODO 3: build the default lazily (only if needed)
    let p2: u16 = parse_port("oops")/* ? */;

    // TODO 4: fall back to u16::default() (= 0)
    let p3: u16 = parse_port("oops")/* ? */;

    // TODO 5: panic with a message if it must succeed
    let p4: u16 = parse_port("oops")/* ? */;
}
```



# Debugging

---

Find the bug



The compiler is your **first** debugger - then reach for a print.

```
fn main() {  
    let xs = vec![1, 2, 3];  
    println!("xs = {xs:?}"); // you choose what to show  
    let _n = dbg!(xs.len()) * 2; // prints AND returns the value  
}
```

```
Compiling playground v0.0.1 (/playground)  
Finished `dev` profile [unoptimized + debuginfo] target(s) in 0.49s  
Running `target/debug/playground`
```

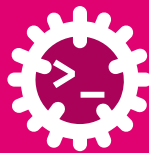
```
[src/main.rs:6:12] xs.len() = 3 ← dbg! prints *file:line*  
                             the *expression* and its *value*
```

```
xs = [1, 2, 3]                ← println! prints what you choose
```

- **println!** - quick, but you **format it yourself**
- **dbg!(expr)** - prints **file:line**, the **expression** and its **value**, then **returns it**
- **rustc --explain E0382** - the **full story** behind an error code



Read the borrow-checker hint first - it usually names the fix.



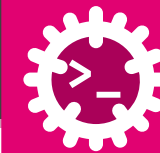
Beyond a quick print: **levels** you can switch on/off **without recompiling**.

```
use log::{info, warn, error};

fn main() {
    env_logger::init();
    info!("starting on port {}", 8080);
    warn!("connection slow, retrying");
    error!("giving up");
}
```

```
RUST_LOG=info cargo run    # show info and above, everywhere
RUST_LOG=debug cargo run   # show debug and above
RUST_LOG=trace cargo run   # everything
```

- levels: error! > warn! > info! > debug! > trace!
- **log** is a **facade** - pick a backend: **env\_logger**, **tracing**
- **tracing** - structured spans, great for **async** / production
- filter via RUST\_LOG, **no code change**

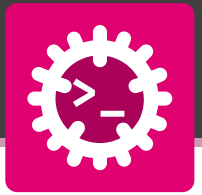


```
===== RUST_LOG=info =====  
[2026-06-23T06:41:25Z INFO logging_levels] application starting  
[2026-06-23T06:41:25Z WARN logging_levels::net] attempt 1 failed, retrying  
[2026-06-23T06:41:25Z WARN logging_levels::net] attempt 2 failed, retrying  
[2026-06-23T06:41:25Z INFO logging_levels::net] connected on attempt 3  
[2026-06-23T06:41:25Z WARN logging_levels] disk almost full: 92% used  
[2026-06-23T06:41:25Z ERROR logging_levels] database connection lost  
[2026-06-23T06:41:25Z INFO logging_levels] application stopped
```

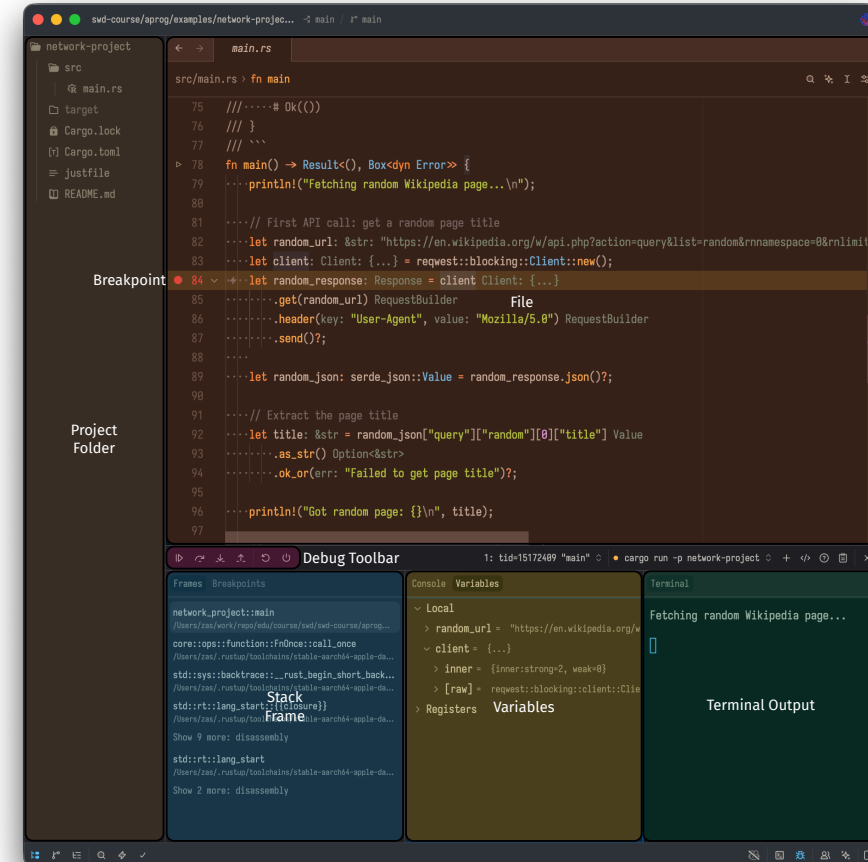
```
===== RUST_LOG=debug =====  
[2026-06-23T06:41:25Z INFO logging_levels] application starting  
[2026-06-23T06:41:25Z DEBUG logging_levels::net] attempt 1/3  
[2026-06-23T06:41:25Z WARN logging_levels::net] attempt 1 failed, retrying  
[2026-06-23T06:41:25Z DEBUG logging_levels::net] attempt 2/3  
[2026-06-23T06:41:25Z WARN logging_levels::net] attempt 2 failed, retrying  
[2026-06-23T06:41:25Z DEBUG logging_levels::net] attempt 3/3  
[2026-06-23T06:41:25Z INFO logging_levels::net] connected on attempt 3  
[2026-06-23T06:41:25Z WARN logging_levels] disk almost full: 92% used  
[2026-06-23T06:41:25Z ERROR logging_levels] database connection lost  
[2026-06-23T06:41:25Z INFO logging_levels] application stopped
```

```
===== RUST_LOG=trace =====  
[2026-06-23T06:41:25Z INFO logging_levels] application starting  
[2026-06-23T06:41:25Z TRACE logging_levels::net] entering connect()  
[2026-06-23T06:41:25Z DEBUG logging_levels::net] attempt 1/3  
[2026-06-23T06:41:25Z WARN logging_levels::net] attempt 1 failed, retrying  
[2026-06-23T06:41:25Z DEBUG logging_levels::net] attempt 2/3  
[2026-06-23T06:41:25Z WARN logging_levels::net] attempt 2 failed, retrying  
[2026-06-23T06:41:25Z DEBUG logging_levels::net] attempt 3/3  
[2026-06-23T06:41:25Z INFO logging_levels::net] connected on attempt 3  
[2026-06-23T06:41:25Z WARN logging_levels] disk almost full: 92% used  
[2026-06-23T06:41:25Z ERROR logging_levels] database connection lost  
[2026-06-23T06:41:25Z INFO logging_levels] application stopped
```

# Debugging with Breakpoints (in Zed / VS Code)



1. Open the *main.rs* file and click the *gutter* to set a *breakpoint*
2. run the program in *debug mode* F4
3. Go through the program with the following actions:
  - *continue* - F5 - run until the next breakpoint
  - *step over* - F7 - run the current line  
stop at the next line
  - *step into* - Ctrl+F11 - run the current line  
stop at the first line of the called function
  - *step out* - Shift+F11 - run until the current  
function returns
4. **Inspect** live the:
  - *stack frame* - see the **call stack**
  - *variables* - see the **current values** of variables in scope
  - *terminal* - see the **program output**





# Profiling

---

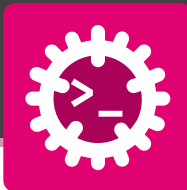
Make it fast







Coming up: five **mystery programs** - guess what it does from the time **split** and the **flamegraph**.



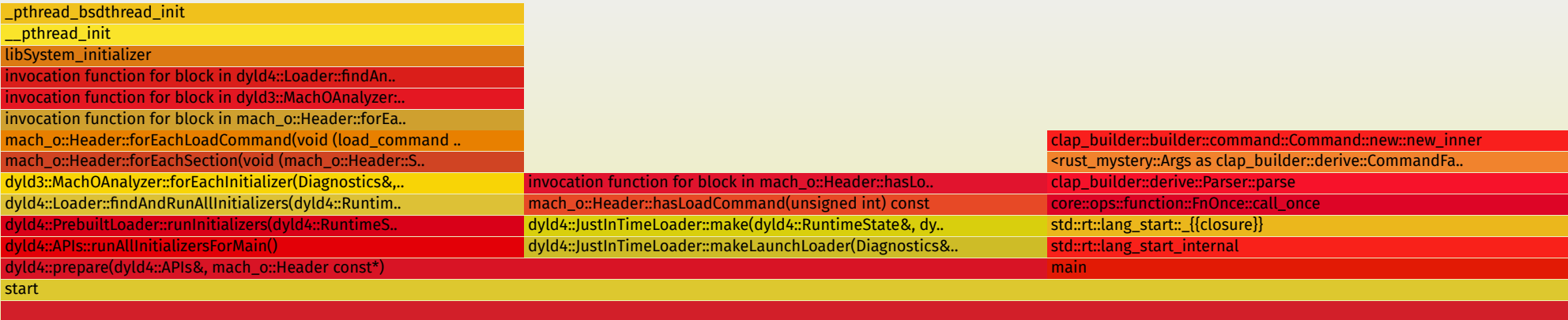
```
hyperfine --warmup 3 --show-output --min-runs 10 "release/rust_mystery_v1_0_0_Mac_AARCH64 -m 1"
```

## Execution Mystery Program 1

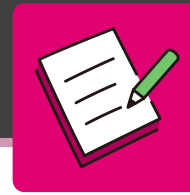
```
...
Time (mean ± σ):      2.013 s ±  0.002 s   [User: 0.003 s, System: 0.004 s]
Range (min ... max):  2.009 s ...  2.014 s   10 runs
```

## Flame Graph

Search



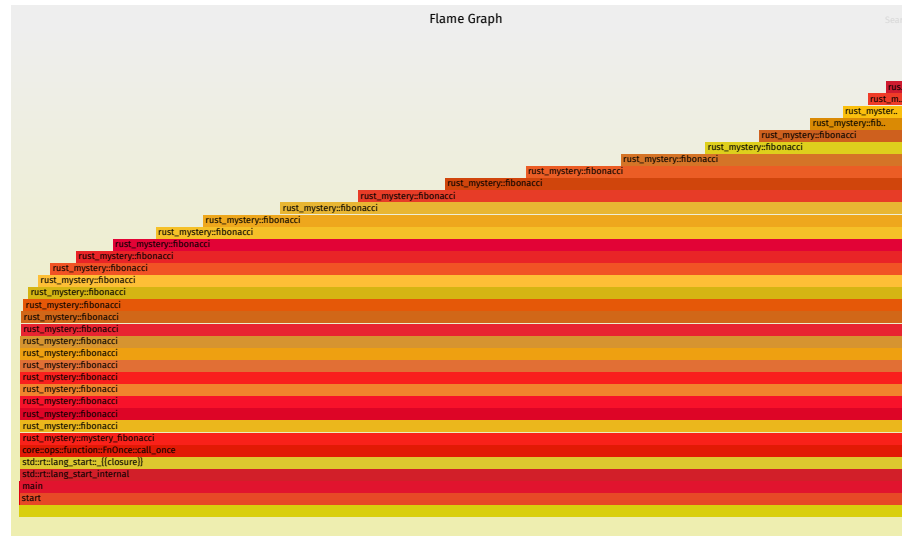
# Mystery Program 2



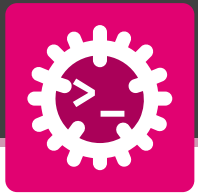
```
hyperfine --warmup 3 --show-output --min-runs 10 "release/rust_mystery_v1_0_0_Mac_AARCH64 -m 2"
```

## Execution Mystery Program 2

```
...  
Time (mean ± σ):      2.213 s ± 0.009 s [User: 2.192 s, System: 0.016 s]  
Range (min ... max):  2.201 s ... 2.227 s 10 runs
```



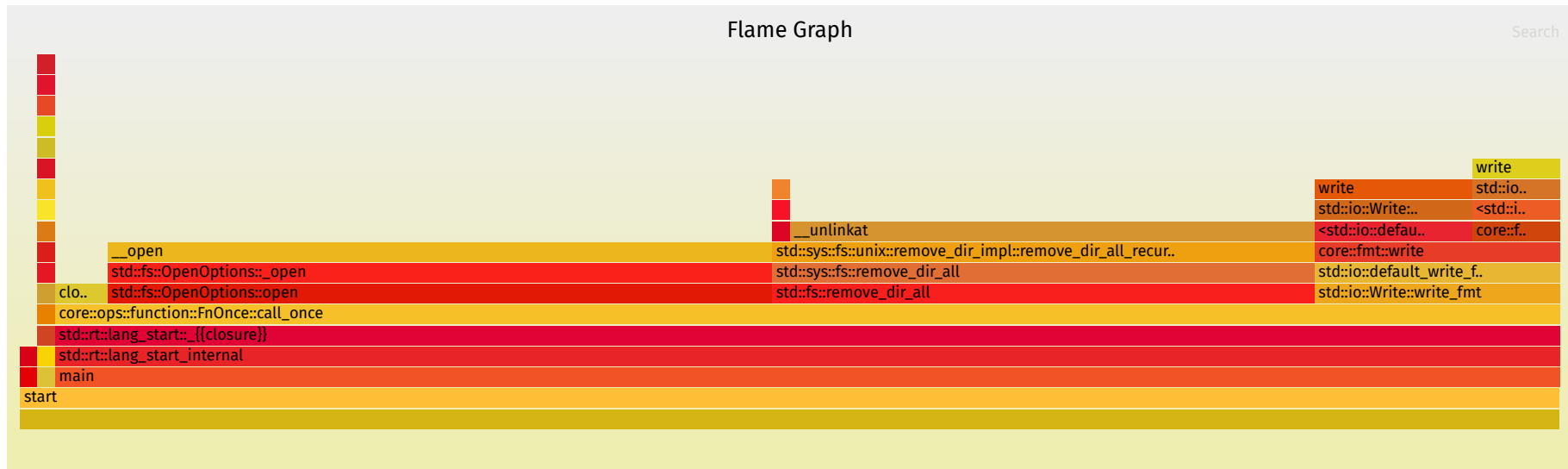
# Mystery Program 3



```
hyperfine --warmup 3 --show-output --min-runs 10 "release/rust_mystery_v1_0_0_Mac_AARCH64 -m 3"
```

## Execution Mystery Program 3

```
...  
Time (mean ± σ):    115.0 ms ± 18.9 ms   [User: 2.5 ms, System: 107.6 ms]  
Range (min ... max): 83.0 ms ... 143.0 ms   24 runs
```





```
hyperfine --warmup 3 --show-output --min-runs 10 "release/rust_mystery_v1_0_0_Mac_AARCH64 -m 4"
```

Execution Mystery Program 4

```
...
Time (mean ± σ):      760.5 ms ± 45.9 ms   [User: 10451.8 ms, System: 41.7 ms]
Range (min ... max):  703.5 ms ... 852.2 ms   10 runs
```

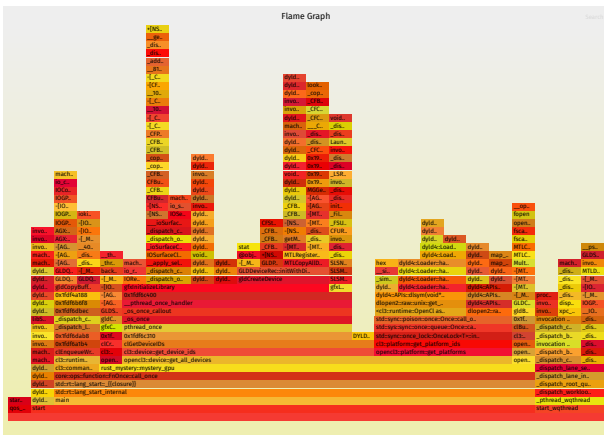
## Flame Graph

Search

```
<.. core::num::_<impl i32>::overflowing_add rust_mystery::mystery_parallel_cpu::_{{closure}}
std::thread::Builder::spawn_unchecked::_{{closure}}::_{{closure}}
std::sys::pal::unix::thread::Thread::new::thread_start
pthread_start
thread_start
```



```
Execution Mystery Program 5
results front: 301
results back: 1300
kernel execution duration (ns): 204
...
Time (mean ± σ):      47.4 ms ±   5.6 ms    [User: 11.0 ms, System: 13.5 ms]
Range (min ... max):  41.0 ms ...  79.5 ms    62 runs
```

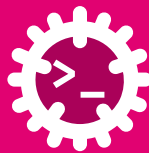




# Documentation

---

Code others (and future-you) can read



Doc comments become a **browsable website** - and the examples are **compiled and tested**.

```
//! # mycrate
//!
//! Temperature conversions for the AProg course.
//! Crate-level docs (`//!`) sit at the top of `lib.rs`
//! and become the front page of the generated site.
//!
//! # Examples
//! ```
//! use mycrate::temperature::Celsius;
//! assert_eq!(Celsius::new(0.0).to_fahrenheit(), 32.0);
//! ```
/// Temperature handling.
pub mod temperature {
    /// A temperature in degrees Celsius.
    ///
    /// # Examples
    /// ```
    /// use mycrate::temperature::Celsius;
    /// let t = Celsius::new(25.0);
    /// assert_eq!(t.to_fahrenheit(), 77.0);
    /// ```
```

```
pub struct Celsius(f64);
impl Celsius {
    /// Creates a new [`Celsius`] value.
    pub fn new(deg: f64) → Self {
        Self(deg)
    }

    /// Converts to Fahrenheit.
    ///
    /// # Panics
    /// Never - this conversion is total.
    pub fn to_fahrenheit(&self) → f64 {
        self.0 * 9.0 / 5.0 + 32.0
    }
}
```





Documentation lives **next to the code** - the same source is both the comments and the published site.

## Writing the comments

- `///` documents the **next item** - a struct, method, or function
- `//!` documents the **enclosing** item - the module or whole crate
- **Markdown** inside: headings, lists, code blocks
- convention sections: `# Examples`, `# Panics`, `# Errors`
- `[Celsius]` - an **intra-doc link** to another item

## Building & testing

- the `# Examples` block is **real code**, compiled and run
- `cargo test` executes every doc example → docs **can't drift**
- public crates publish automatically to **docs.rs**

```
# build the HTML docs, open in browser
cargo doc --open

# also compiles & runs the doc examples
cargo test
```

WHY DO WHALES JUMP  
WHY ARE THERE MIRRORS ABOVE BEDS  
WHY DO I SAY UH  
WHY IS SEA SALT BETTER  
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS  
WHY IS THERE NOT A POKEMON MMO  
WHY IS THERE LAUGHING IN TV SHOWS  
WHY ARE THERE DOORS ON THE FREEWAY  
WHY ARE THERE SO MANY SVCHOST.EXE RUNNING  
WHY AREN'T ANY COUNTRIES IN ANTARCTICA  
WHY ARE THERE SCARY SOUNDS IN MINECRAFT  
WHY IS THERE KICKING IN MY STOMACH  
WHY ARE THERE TWO SLASHES AFTER HTTP  
WHY ARE THERE CELEBRITIES  
WHY DO SNAKES EXIST  
WHY DO OYSTERS HAVE PEARLS  
WHY ARE DUCKS CALLED DUCKS  
WHY DO THEY CALL IT THE CLAP  
WHY ARE KYLE AND CARTMAN FRIENDS  
WHY IS THERE AN ARROW ON AANG'S HEAD  
WHY ARE TEXT MESSAGES BLUE  
WHY ARE THERE MUSTACHES ON CLOTHES  
WHY WUBA LUBBA DUB DUB MEANING  
WHY IS THERE A WHALE AND A POT FALLING  
WHY ARE THERE SO MANY BIRDS IN SWISS  
WHY IS THERE SO LITTLE RAIN IN WALLIS  
WHY IS WALLIS WEATHER FORECAST ALWAYS WRONG

WHY ARE THERE MALE AND FEMALE BIKES  
WHY ARE THERE BRIDESMAIDS  
WHY DO DYING PEOPLE REACH UP  
HOW FAST IS LIGHTSPEED  
WHY ARE OLD KLINGONS DIFFERENT

WHY ARE THERE  
SQUIRRELS



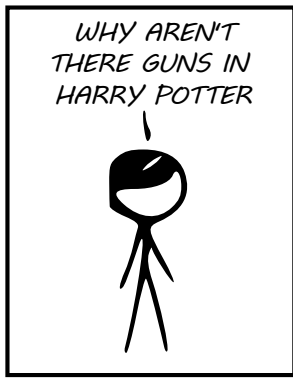
WHY AREN'T THERE DINOSAUR GHOSTS  
WHY DO IGUANAS DIE  
WHY ARE HATS SO EXPENSIVE  
WHY IS THERE CAFFEINE IN MY SHAMPOO  
WHY HAVE DINOSAURS NO FUR  
WHY AREN'T ECONOMISTS RICH  
WHY DO AMERICANS CALL IT SOCCER  
WHY ARE MY EARS RINGING  
WHY IS 42 THE ANSWER TO EVERYTHING  
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER  
WHY IS MARVIN ALWAYS SO SAD

WHY ARE THERE TINY SPIDERS IN MY HOUSE  
WHY DO SPIDERS COME INSIDE  
WHY ARE THERE HUGE SPIDERS IN MY HOUSE  
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE  
WHY ARE THERE SPIDERS IN MY ROOM  
WHY ARE THERE SO MANY SPIDERS IN MY ROOM  
WHY DO SPYDER BITES ITCH

WHY ARE THERE ANTS  
WHY IS EARTH TILTED  
WHY IS SPACE BLACK  
WHY IS OUTER SPACE SO COLD  
WHY ARE THERE PYRAMIDS ON THE MOON  
WHY IS NASA SHUTTING DOWN

WHY ARE THERE FEMALE  
WHY AREN'T THERE GUNS IN HARRY POTTER

WHY ARE THERE FEMALE



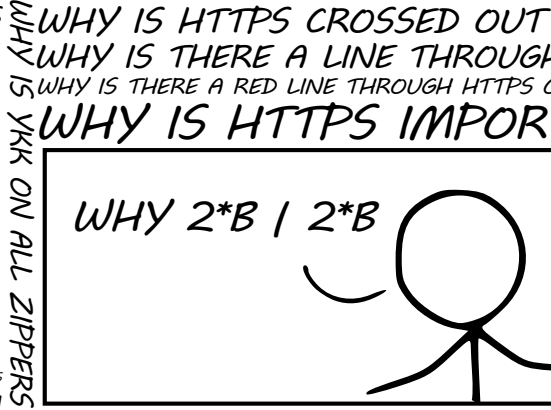
WHY ARE SWISS AFRAID OF DRAGONS  
WHY IS HTTPS CROSSED OUT  
WHY IS THERE A LINE THROUGH  
WHY IS THERE A RED LINE THROUGH HTTPS  
WHY IS HTTPS IMPOR

# QUESTIONS

CAN BE ASKED BY ANYONE ANYTIME

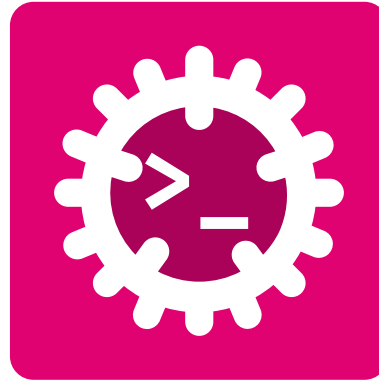
WHY IS THERE A SWARM OF ANTS  
WHY IS THERE PILGRIM  
WHY ARE THERE SO MANY CROWS IN ROCHESTER  
WHY IS TO BE OR NOT TO BE FL  
WHY DO CHILDREN GET CANCER  
WHY IS POSEIDON ANGRY WITH ODYSSEUS  
WHY IS THERE ICE IN SPACE

WHY IS THERE AN OWL IN MY BACKYARD  
WHY IS THERE AN OWL OUTSIDE MY WINDOW  
WHY IS THERE AN OWL ON THE DOLLAR BILL  
WHY DO OWLS ATTACK PEOPLE  
WHY ARE FPGA's EVERYWHERE  
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE  
WHY ARE THERE GODS  
WHY ARE THERE TWO SPOCKS



WHY ARE MY BOOBS ITCHY  
WHY ARE CIGARETTES LEGAL  
WHY ARE THERE DUCKS IN MY POOL  
WHY IS JESUS WHITE  
WHY IS THERE LIQUID IN MY EAR  
WHY DO Q TIPS FEEL GOOD  
WHY DO PEOPLE DIE

WHAT IS <https://xkcd.com/1256/>  
WHY DO THEY SAY T-MINUS  
WHY ARE OCEANS BECOMMING MORE ACIDIC



## Glossary & Bibliography

---



## Computer Science

**CPU – Central Processing Unit:** The main processing component of a computer. [23](#)

## Concurrency

**I/O – Input/Output:** Operations that involve reading from or writing to external devices or resources. [23](#)



## Bibliography

- [1] S. Klabnik, C. Nichols, and C. Kryocho, “The Rust Programming Language.”
- [2] J. Blandy, J. Orendorff, and L. F. s Tindall, *Programming Rust: Fast, Safe Systems Development*. O'Reilly Media, Incorporated, 2021.
- [3] H. Collard, *Black Hat Rust*, vol. 1. Amazon, 2025.
- [4] A. Shvets, “Refactoring Guru.” 2026.
- [5] R. C. Martin, *Clean Code: A Handbook of Agile Software Craftsmanship*. Pearson Education, 2008.
- [6] “SVG Repo - Free SVG Vectors and Icons.”